

Computer Vision

Tawei Tseng

2. The Fundamental Matrix

Fundamental matrix:

$$F = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

Epipolar line in image 2 for $x = (0,1,1)^\top$:

$\ell' = Fx = (-1, 0, 0)^\top$, i.e. the line $x' = 0$.

Among the candidate image-2 points, only $(0,1)$ lies on the line $x' = 0$, so only $(0,1)$ can be a projection of the same 3D point X into P_2 .

Epipoles:

Camera center of $P_1=[I \ 0]$ is $C_1 = (0,0,0,1)^\top$.

Projection into image 2: $e_2 = P_2 C_1 = (2,1,1)^\top$.

Camera center of P_2 is the right nullspace of P_2 : solving $P_2 C_2 = 0$ gives $C_2 = (-1,1,-1,1)^\top$.

Projection into image 1: $e_1 = P_1 C_2 = (-1,1,-1)^\top$.

Fundamental matrix (using $P_1=[I \ 0]$ so $F = [e_2]_\times P_2$):

$$F = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -2 \\ -1 & 2 & 0 \end{pmatrix}$$

Determinant: $\det(F) = 0$.

Verification of epipole properties:

$$e_2^T F = (2, 1, 1) \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -2 \\ -1 & 2 & 0 \end{pmatrix} = (0, 0, 0).$$

$$F e_1 = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -2 \\ -1 & 2 & 0 \end{pmatrix} (-1, 1, -1)^T = (0, 0, 0)^T.$$

F for the original (unnormalized) points is

$$F = N_2^T \tilde{F} N_1$$

$$x_{1X} = (0, 3, 1)^T, \quad x_{2X} = (-5, 8, -3)^T,$$

$x_{1Y} = (-1, 2, 0)^T, \quad x_{2Y} = (1, 4, 3)^T$; all four satisfy $x_2^T F x_1 = 0$, and the camera center of P_2 is $C_2 = (-2, -1, 1, 0)^T$.

With $P_1 = [I \ 0]$, a 3D point $X = (0, 3, 1, 1)^T$ projects to $x_{1X} = P_1 X = (0, 3, 1)^T$, and $Y = (-1, 2, 0, 1)^T$ projects to $x_{1Y} = P_1 Y = (-1, 2, 0)^T$.

The epipole in image 2 is the right null vector of F^T , $e_2 = (-1, 0, 1)^T$

$$\text{so } [e_2]_{\times} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \text{ and } P_2 = [[e_2]_{\times} F e_2] = \begin{pmatrix} -2 & 0 & -4 & -1 \\ 0 & 2 & 2 & 0 \\ -2 & 0 & -4 & 1 \end{pmatrix}$$

Then $x_{2X} = P_2 X = (-5, 8, -3)^T$ and $x_{2Y} = P_2 Y = (1, 4, 3)^T$.

For point X , the scalar $x_{2X}^T F x_{1X}$ evaluates to 0, so the two projections obey the epipolar constraint. For point Y , the same computation gives $x_{2Y}^T F x_{1Y} = 0$, so its projections also lie on corresponding epipolar lines.

The camera center C_2 is the (projective) point satisfying $P_2 C_2 = 0$, i.e. the right nullspace of P_2 .

Solving yields $C_2 = (-2, -1, 1, 0)^T$, a point at infinity indicating that this camera is affine.

3. The Essential Matrix

Using the given U, V one finds $U^T V$ is a 3×3 orthogonal matrix with $\det(U^T V) = 1$.

Compute

$$E = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} V^T = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} V^T = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \end{pmatrix}.$$

Take the homogeneous image points $x_1 = (1, 0) \mapsto \tilde{x}_1 = [1, 0, 1]^T$ (camera 1) and $x_2 = (0, 1) \mapsto \tilde{x}_2 = [0, 1, 1]^T$ (camera 2). Check the epipolar constraint

$$\tilde{x}_2^T E \tilde{x}_1 = [0 \ 1 \ 1] E [1 \ 0 \ 1]^T = 0.$$

So (x_1, x_2) is a plausible correspondence.

Projection by $P_1 = [I \ 0]$ gives camera-1 coordinates (X_1, X_2, X_3) . The image $x_1 = (1, 0)$ means

$$\frac{X_1}{X_3} = 1, \frac{X_2}{X_3} = 0 \Rightarrow X_1 = X_3, X_2 = 0.$$

A convenient homogeneous representative is

$$\mathbf{X}(s) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ s \end{pmatrix}$$

For each candidate $P_2 = [R \ t]$ we require that $\mathbf{X}(s)$ projects to $x_2 = (0, 1)$.

Projection in camera 2 gives the homogeneous 3-vector $P_2 \mathbf{X}(s)$. The image coordinates (inhomogeneous) are the first two components divided by the third, but remember these are *not* Cartesian camera coordinates unless we divide by the homogeneous coordinate s . Equivalently, the Cartesian camera-2 coordinates are $(P_2 \mathbf{X}(s))/s$. Enforcing the image $x_2 = (0, 1)$

Solving the two scalar equations for s for each of the four choices gives:

For $P_2 = [UWV^T, +u_3]$ one obtains $s = +\sqrt{2}$.

For $P_2 = [UWV^T, -u_3]$ one obtains $s = -\sqrt{2}$.

For $P_2 = [UW^TV^T, +u_3]$ one obtains $s = -\sqrt{2}$.

For $P_2 = [UW^TV^T, -u_3]$ one obtains $s = +\sqrt{2}$.

A finite 3D point is in front of a camera if its **Cartesian** z -coordinate in that camera

frame is positive. With $\mathbf{X}(s) = \begin{pmatrix} 1 \\ 0 \\ 1 \\ s \end{pmatrix}$ the Cartesian depth in camera 1 is

$$z_1 = \frac{X_3}{X_4} = \frac{1}{s},$$

and the Cartesian depth in camera 2 is

$$z_2 = \frac{(P_2 \mathbf{X}(s))_3}{s}.$$

Evaluating the depths for the four solutions gives:

$P_2 = [UWV^T, +u_3]$ with $s = +\sqrt{2}$:

$$z_1 = \frac{1}{\sqrt{2}} > 0, z_2 = 1 > 0.$$

point is in front of both cameras. **VALID**.

$P_2 = [UWV^T, -u_3]$ with $s = -\sqrt{2}$:

$$z_1 = \frac{1}{-\sqrt{2}} < 0, z_2 = -1 < 0.$$

behind both cameras. **INVALID**.

$P_2 = [UW^TV^T, +u_3]$ with $s = -\sqrt{2}$:

$$z_1 < 0, z_2 > 0.$$

behind camera 1. **INVALID.**

$P_2 = [UW^T V^T, -u_3]$ with $s = +\sqrt{2}$:

$$z_1 > 0, z_2 < 0.$$

behind camera 2. **INVALID.**

Thus the only physically valid choice (point in front of both cameras) is

$P_2 = [UWV^T, u_3]$, with $s = +\sqrt{2}$, and the corresponding 3D homogeneous point is

$\mathbf{X}(\sqrt{2}) = (1, 0, 1, \sqrt{2})^T$. Its Cartesian coordinates in camera 1 are $(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$ and in

camera 2 are $(0, 1, 1)$